Suvir Shah

Bachelor of Technology
Computer Science and Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., CSE	National Institute of Technology, Patna	7.48/10	2028
Senior Secondary	St. Michael's High School	90.3%	2022
Secondary	St. Michael's High School	89.3%	2020

Projects

- **CodeArena – Full-Stack Coding Practice Platform** *2026*
  Tech: React, Redux, Node.js, Express.js, MongoDB, Tailwind CSS, Judge0 API, JWT, Redis, Cloudinary, Monaco Editor []
  - Developed a LeetCode-style coding platform supporting 3 programming languages (C++, Java, JavaScript) with JWT-based authentication, Redis session management, and real-time code execution via Judge0 API
  - Built role-based admin console handling 100+ coding problems with CRUD operations, editorial video uploads to Cloudinary, and submission tracking for 500+ user submissions
  - Designed 4 MongoDB schemas (User, Problem, Submission, Video) storing 10,000+ data records with optimized queries reducing response time by 40%
  - Created interactive DSA visualizer with 15+ algorithm animations (Bubble Sort, Merge Sort, Binary Search, DFS, BFS, etc.) covering 8 data structures (Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Hash Maps, Heaps)
- **Programming Lecture Video RAG Pipeline** *2026*
  Tools: Python, Whisper (large-v2), Ollama (bge-m3), scikit-learn, Pandas/Joblib, FFmpeg []
  - Converted 10+ CP lecture videos (10+ hours) to MP3 using FFmpeg, transcribed Hindi content into 2,000+ timestamped JSON chunks (avg. 30 seconds each) with Whisper large-v2 achieving 95% accuracy
  - Generated bge-m3 embeddings for all 2,000+ chunks via Ollama API, creating scalable Pandas/joblib index with 50MB storage footprint
  - **Developed** cosine similarity retriever with **semantic search** returning top-5 relevant chunks with video/timestamp precision (avg. 2-second accuracy) for **retrieval-augmented generation (RAG)** LLM prompting
  - CLI answers "which video has [solution]" queries with precise minute:second guidance, reducing manual video search time from 15 minutes to under 30 seconds
- **House Price Prediction using Machine Learning** *Nov 2024 – Dec 2024*
  Tools: Python, Scikit-learn, Pandas, NumPy []
  - Engineered end-to-end regression pipeline on California housing dataset (20,640 samples, 8 features) using RandomForestRegressor with 87% R^2 score and 0.025 MAE
  - Serialized trained model (15MB) and preprocessing pipeline enabling 10ms inference time on new housing data without retraining
  - Visualized feature importance ranking (top 3: Median Income: 42%, House Age: 18%, Occupancy: 12%) and error distribution (RMSE: $48,500) to identify key price drivers

Skills

- **Programming Languages**: C++17, Python 3.11, JavaScript, SQL, Java
- **Technologies**: React/Next.js, Node/Express, MongoDB, Redis, JWT, PyTorch, scikit-learn, Whisper, Ollama (bge-m3), Tailwind CSS, REST APIs
- **Tools**: FFmpeg, Pandas/Joblib, Git/GitHub, VS Code, LeetCode/Codeforces, Vercel, Parcel

Certifications

- **Code With Harry**, The Ultimate Job Ready Data Science Course *Feb 2026*

Achievements

- Codeforces: 1000+rating & CodeChef: 2 Stars
- LeetCode: 490+ Problems & GeeksforGeeks: 380+ problems solved
- JEE Mains: 99 percentile